Maths

Probability

1. Key concepts in probability theory:
    1. Let's say we have two RV X and Y:
        1. Joint distribution: $p(X, Y)$ (reads "X and Y")
        2. Marginal distribution: p(X) and p(Y)
        3. Conditional distribution: p(X|Y) and p(Y|X) (reads "X given Y" and "Y given X")
        4. Bayes Theorem: $p(X|Y) = \frac{p(X,Y)}{p(Y)} = \frac{p(Y|X)p(X)}{p(Y)}$
            1. Prior: p(X) (what I originally believe about X)
            2. Posterior: p(X|Y) (what I believe about X now that I've seen/observed Y).
        5. Independence: Knowing X tells you nothing about Y
            1. $p(X|Y) = p(X)$ and $p(X, Y) = p(X)p(Y)$.
    2. The goal of probabilistic modeling is to learn the probability distributions
    3. We can usually describe the probability distributions through some **parameters** (θ)
        1. Gaussian: mean(μ) and variance(σ^2)
        2. Poisson: average rate (λ)
        3. Some complicated distribution: can be parametrized by neural networks (θ)
    4. The goal now is to learn those parameters given data
    5. We call the probability of data given model parameters as **likelihood** $p(x|\theta)$. i.e. "if my parameters are accurate, how likely is the data that I observe".
        1. θ is model parameter.
        2. For example RV x is binomial, x is the number of times a coin landed heads in N trials.
            1. $$P(x = X|\theta = p) = \binom{N}{x} p^x (1-p)^{N-x} \quad \text{(likelihood)}$$
        3. if RV x is single variable Gaussian, for modeling a real value, e.g. temperature sensor reading.
            1. $$P(x = X|\theta = (\mu, \sigma^2)) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \quad \text{(likelihood)}$$
        4. Note: Notice that in both cases the RV x ended up in the exponent.
        5. Bayes rule:

$$p(\theta|x = D) = \frac{p(x|\theta)p(\theta)}{p(x)}$$

$p(x|\theta)$ is the likelihood, $p(\theta)$ is the prior distribution, $p(\theta|x = D)$ is the posterior, and $p(x)$ is the normalization term.

2. Probabilistic graphical models can only do 2 things [4]:
    1. Inference: calculating a probability of some data modelled as RV using the model $\mathcal{M}$:
        1. evaluation $P(obs|state)$,
        2. decoding $P(state|obs)$
    2. Learning model parameters: refers to finding the graphical model $\mathcal{M}$, given some data $\mathcal{D}$.
        1. MAP (maximum a posteriori)
        2. Maximum likelihood
        3. expectation maximization
        4. forward-backward algorithm
3. Conditional probability can be computed using:
    1. Bayes rule
    2. Enumeration

Probabilistic Inference Methods [5]

1. Maximum Likelihood Estimation (MLE)
    1. $x^* = \arg\max p(d|x)$.

2. Likelihood: $p(d|x)$
3. $x*$ is the most probable value of x.
4. $\{x\}$ is a set of possible values of x (the candidates of x).
5. $\arg\max$ to return the value of x that maximizes $p(d|x)$.
6. The weakness of MLE is that it allows impossible candidates, as it treats candidates uniformly.
7. The intuitive meaning of MLE:
    1. Take each possible illness e.g. x=flu.
    2. Verify if it fits to the symptoms $\rightarrow$ probability of x.
    3. Decide the illness based on the hights probability.
2. Maximum A Posteriori (MAP)
    1. $$\begin{aligned} x^* &= \arg\max p(x|d) \\ &= \arg\max p(d|x)p(x) \end{aligned}$$

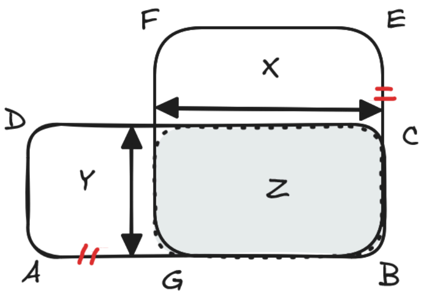
    . Likelihood:
    2. MAP treats the candidates differently: $p(x = \text{flu}) > p(x = \text{cancer})$
    3. Main criticism of MAP is how to determine $p(x)$.
3. Full Bayesian Inference
    1. $p(x|d) = \frac{p(d|x)p(x)}{p(d)}$. $P(d)$ is intractable to solve.
    2. This is the ideal and most robust inference that besides providing a prediction on the value of x, it also provides the uncertainty, or how high/low the probability of the prediction.
    3. It's possible because of $p(d)$, the evidence. This will normalize the product of $p(d|x)p(x)$.
    4. However to obtain $p(d)$ is analytically intractable, and in discrete, it can be prohibitively expensive.
4. Example: According to our observation, a patient has a set of symptoms: d : {stiff neck, coughing, headache, fever}. We also have a set of possible illness: flu, meningitis, infection and cancer.

Lagrange Multiplier [3]

Hungarian Matching [2]

AM, GM [8]

1. Arithmetic mean and the Geometric mean
    1. Arithmetic mean
        1. A careful experimenter measures a certain object and finds its length to be 2.172 feet. On repeating his measurements twice more he obtains the lengths 2.176 ft and 2.171 ft. What should he accept as the true length of the object? In such a case it is customary to use the average measurements, to add them and divide by their number.
        2. The experimenter would find the total to be 6.519 and dividing by 3 yields 2.173 ft. as the length of the object. This average that we have described is called the arithmetic mean.
        3. The arithmetic mean of n numbers $a_1, a_2, a_3, \ldots, a_n$ is $AM = \frac{1}{n}(a_1 + a_2 + a_3 + \ldots + a_n)$.
        4.
    2. Geometric mean
    1. An object is to be weighted by putting it on one pan of a balance and **counterbalancing it with weights** on the other pan.
    2. We shall make two weighings, one with the object in the left pan, one with it on the right pan. If the results of these two weighings are a_1 and a_2, what should we take as the true weight?
    3. If w is the true weight of the obj this means $wl = a_1 r$ for the first weighing and $wr = a_2 l$ for the second.
    4. $wlwr = a_1 r a_2 l$, or dividing by lr, $w^2 = a_1 a_2$. The true weight of the object is given by the formula $w = \sqrt{a_1 a_2}$. This is a new sort of average. It is called the geometric mean of a_1 and a_2.
    5. The geometric mean of the n positive numbers $a_1, a_2, a_3, \ldots, a_n$ is $G = \sqrt[n]{a_1 a_2 a_3 \ldots a_n}$.
    3. **The square is larger than all other rectangles with the same perimeter**
        1. rectangle ABCD has a fixed perimeter P. The square BEFG has each side a length of $\frac{1}{4}P$, so its perimeter is also P.

        2. 

        3. In the figure the rectangle Z is part of both the original rectangle and the square. The square also contains the area X, while the original rectangle is made up of Z and Y.
        4. Now, AB+BC is half the perimeter of the rectangle, and GB+BE is half the perimeter of the square, so these two are equal; AB+BC=GB+BE. This can be written as AG+GB+BC=GB+BC+CE, from which we have AG=CE.
        5. Thus the rectangle X is just as high as Y is wide. However, the width of X is one side of the square, while the height of Y is *part* of the side of the square, and is therefore smaller.
        6. If two rectangles are the same length in one dimension, then the one with the larger other dimension has the larger area, i.e. area(X)>area(Y).
        7. Therefore X+Z>Y+Z, and the square is of larger area than the rectangle.
        8. The height of Y would no longer be a part of the side of the square only if ABCD were a square itself, and then the square BEFG would itself be the original rectangle ABCD (but vertical).
        9. Therefore the square is larger than all other rectangles with the same perimeter.
    4. **The geometric mean of two numbers is always less than their arithmetic mean**. The two are equal only if x and y are equal.
        1. Proof: Let x and y be the dimensions of rectangle in inches.
        2. Next, following from the proof above, that a square is larger than all other rectangles of the *same perimeter**.
        3. The are of the rectangle may then be given in square inches and its perimeter is $x+y+x+y=2(x+y)$ inches. Therefore the square has sides $\frac{1}{2}(x+y)$ inches in length.
        4. Thus if x and y are any two positive numbers, the results becomes: $xy \leq \left(\frac{x+y}{2}\right)^2$.
    5. The geometric mean of n positive numbers is never larger than the arithmetic mean, $GM \leq AM$.
        1. Proof by Cauchy:
2. Arithmetic mean and the root mean square.
    1. The arithmetic mean is not larger than the root mean square.
    2. $AM \leq RMS$

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